Project ARENA

Boobook-DAA Test Plan

Contract AFHQ/JDI/14/25

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Revision	Date	Author	Change Description
1.0	12/02/2026	David Johnson	Derived from MS0052 and revised for Prototype D (Boobook-DAA)
2.0	26/02/2026	David Johnson	Updated with more detailed verification test procedures in response to JDI feedback.

List of Acronyms

ACAS	Airborne Collision Avoidance System
ACMA	Australian Communications and Media Authority
ARPANSA	Australian Radiation Protection and Nuclear Safety Agency
ASCA	Australian Strategic Capability Accelerator
ATAR	Air To Air Radar
DAA	Detect And Avoid
DIH	Defence Innovation Hub
DRePL	Defence Remote Pilot Licence
DWC	DAA Well Clear
EO	Electro-optic
FM	Frequency Modulated
GPS	Global Positioning System
IR	Infra-red
JDI	Jericho Disruptive Innovation
MRR	Measurement Repetition Rate
PDB	Power Distribution Board
POH	Pilot Operators Handbook
REOC	Remote Operator Certificate
RTCA	Radio Technical Commission for Aeronautics
SAR	Synthetic Aperture Radar
SOP	Standard Operating Procedure
SSD	System Specification Document
SUT	System Under Test
TPA	Third Party Authorisation
VTOL	Vertical Take-Off and Landing

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1 Introduction

This document describes the field-demonstration test plan for the Boobook-DAA radar modules developed under Jericho Disruptive Innovation (JDI) contract AFHQ/JDI/14/25.

This document is derived prom previous test plans developed under the Defence Innovation Hub (DIH) for tinySAR prototypes B and C, optimised for DAA and SAR functionality respectively.

1.1 tinySAR Prototype D Test Objectives and Outcomes

tinySAR Prototype B was an intermediate prototype to facilitate further development of the system prior to formal testing at the conclusion of the project. The objectives of the tests described in the initial revision of this document were therefore to:

1. Provide evidence of progress towards meeting requirements specified in the system design documentation.
2. Acquire radar and ground-truth data from an aircraft (in-flight) for subsequent post-processing, in order to demonstrate *detect-and-avoid* (DAA) and *synthetic-aperture radar* (SAR) imaging functionality.
3. Inform development of the final System Prototype C.

In reality, Prototype B did reach the required level of performance to make a full suite of flight tests worthwhile, and hence the Prototype B Test Report (MS0072) placed more emphasis on the performance of individual subsystems in order to make improvements towards Prototype C, rather than the original objectives laid out in v0.1 of this document

1.2 tinySAR Prototype C Test Objectives

The objectives of tinySAR Prototype C were then to fully test the performance requirements of the system within the contest of the DIH/ASCA project, noting that some performance metrics must either be tested in isolation, or be altered slightly in order to meet spectrum regulatory requirements.

An additional point to note, is that due to manufacturing difficulties with the conformal antenna of Prototype B (which whilst not thought to be intractable presented a high degree of schedule-risk relative to the limited time remaining for the project), the decision was made that the Prototype C antenna should be constructed in a planar, rather than conformal, antenna configuration, more suited to SAR than DAA measurements.

Based on discussions with potential platform OEMs such as AMSL Aero, Boeing Insitu and Wisk Aerospace, it was also determined that whilst multi-mode operation is useful, one modality is likely to be more important than the other for different applications and platforms. For larger platforms, this is likely to be the DAA mode, whilst for the Insitu Integrator (which has its own optical DAA system and little space in the front of the aircraft, a lightweight SAR module is of greater utility. Having two variants with common electronics and processing, but which differ in antenna configuration, is therefore likely to be of benefit moving forwards.

Whilst we may occasionally refer to tinySAR in this document, this has therefore been changed to something better suited to two variants, namely Boobook-SAR and Boobook-DAA.

1.3 Boobook-DAA Test Objectives

The purpose of the latest round of testing is to ensure that failings in the previous iteration of the system have been successfully corrected and perform air-to-air target detection measurements online. This will then provide sufficient confidence in the radar to take it to the next stage, integrating with the DAA processor developed by Revolution Aerospace for integration on the Camel Train platform.

2 Relevant Functional and Performance Specifications

A full list of specifications can be found in the *System Specification Document* (SSD), of which the most important *test performance metrics* (TPMs) can then be derived as follows:

Table 1 – High-level TPMs for Boobook-DAA

TPM ID	TPM Description	Upper Threshold	Lower Threshold	Target Value	Verification Measure
TPM11.2.13A	Minimum Frequency of Operation (Civil/DAA)	-	15.3GHz	15.3GHz	Spectrum Analyser
TPM11.2.13B	Maximum Frequency of Operation (Civil/DAA)	15.7GHz	-	15.7GHz	Spectrum Analyser
TPM11.2.14A	Minimum Frequency of Operation (Mil/SAR)	-	15.7GHz	15.7GHz	Spectrum Analyser
TPM11.2.14B	Maximum Frequency of Operation (Mil/SAR)	17.2GHz	-	17.2GHz	Spectrum Analyser
TPM2.5.1A	System Power (max load) per module	40W	100W	60W	Benchtop Power Supply
TPM6.1.2B	Module weight#	0.65kg	0.85kg	0.75kg	Scales
TPM6.1.3B	Depth (per module)	100mm	30mm	90mm	Callipers
TPM-11.2.15D	Minimum range (single module)	300m	670m	500m	Test data with ground-truth
TPM11.2.3A	Maximum range (single module)	8.9km	5km	8.5km	Test data with ground-truth
TPM12.5.1A	DAA target tracking error in Azimuth	1deg	2deg	1.5deg	Test data with ground-truth
TPM12.5.1B	DAA target tracking error in elevation	1deg	2deg	1.5deg	Test data with ground-truth
TPM-11.2.16D	DAA-mode azimuth field of regard (two modules)	+/-65deg	+/-55deg	+/-60deg	Test data with ground-truth
TPM-11.2.17D	DAA-mode elevation field of regard (two modules)	+/-25deg	+/-15deg	+/-20deg	Test data with ground-truth
TPM-11.2.18D	DAA-mode target tracking error in range	4m	15m	6m	Test data with ground-truth
TPM-11.2.19D	DAA-mode target tracking error in range-rate	1m/s	2.4m/s	2m/s	Test data with ground-truth

Without cables.

Update v0.3, The verification test procedure for those elements of Table 1 that cannot be identified though direct measurement are now described in further detail in Section 6.

3 Pre-requisite Resources

3.1 System Under Test

The *System Under Test* (SUT) will be two radar modules, which are themselves comprised of many hardware and software subsystems, each subject to a series of internal unit tests and progressively more complex integration tests prior to the full system-testing described here.

The image depicted in Figure 1 highlights the complexity of the Boobook-DAA electronic system, now made up of 21 custom circuit boards (including a patent-pending antenna structure), bespoke *monolithic microwave integrated circuits* (MMICs) and *commercial off the shelf* (COTS) processor, storage and display modules.

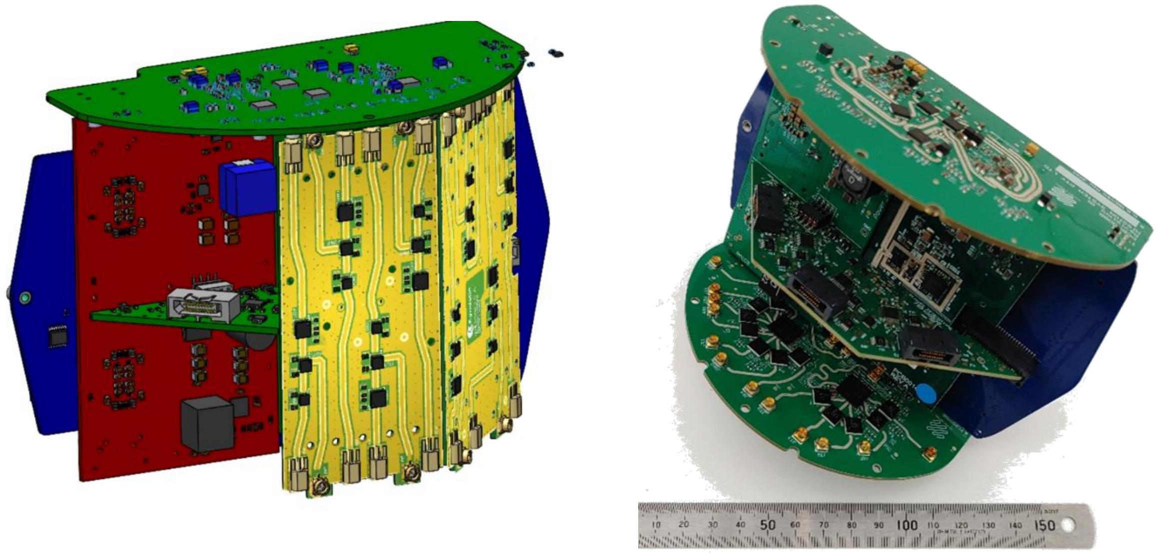

Figure 1 – Representative cutaways of Boobook-DAA electronics

Intended for airborne operation, the radar system is specifically designed to minimise close-in returns from dynamic clutter including propellors, rotors and other moving surfaces. Both for this reason, and because our Third Party Authorisation (TPA) to operate from the Defence Spectrum Office (DSO) will likely require it, integrated testing will take place on board a remotely piloted aircraft in the vicinity of Cultana Military Training Area (SA).

Two radar modules will therefore be installed in a 'Radar Payload' mounted under a T-drones M1200 Uncrewed Air System (UAS), along with sufficient additional sensors, processing, navigation and communications modules to allow safe operation.

The radar modules built for these tests will be the 4th iteration of the tinySAR hardware designed and built by Mission Systems, referred to internally as Prototype D, and externally as Boobook-DAA.

The configuration of the Aerial Test-bed on which a radar payload consisting of two such modules (and associated hardware) can be seen in Figure 2.

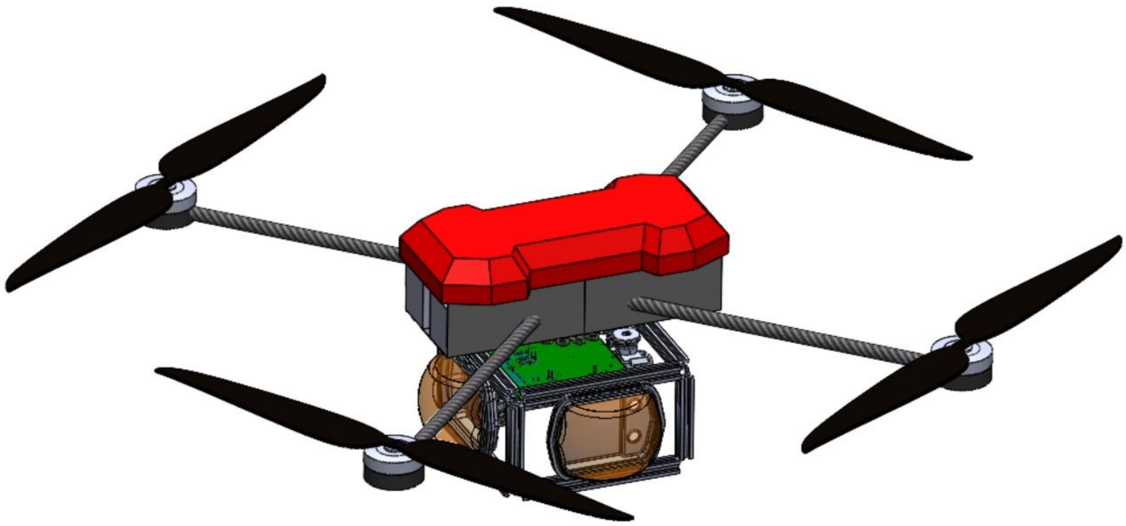

Figure 2 – Radar payload reconfigured for Boobook DAA

3.2 Boobook-DAA operation

A single (Prototype D) DAA module is designed to operate over a sector of airspace (120˚ azimuth, 40˚ elevation), providing detection of discrete targets in that volume using a combination of delayed and interrupted *frequency-modulated* (FM) waveforms to determine a target's location in range and doppler-velocity, and amplitude-comparison monopulse techniques to determine its location in angle.

Figure 3 below shows the method by which targets (in this case synthetic ones) can be detected in azimuth across the antenna, by comparing signals across pairs of subarrays. In this figure, the length of each line is proportional to measurement error, noting that artefacts appearing at close range are due to a display issue.

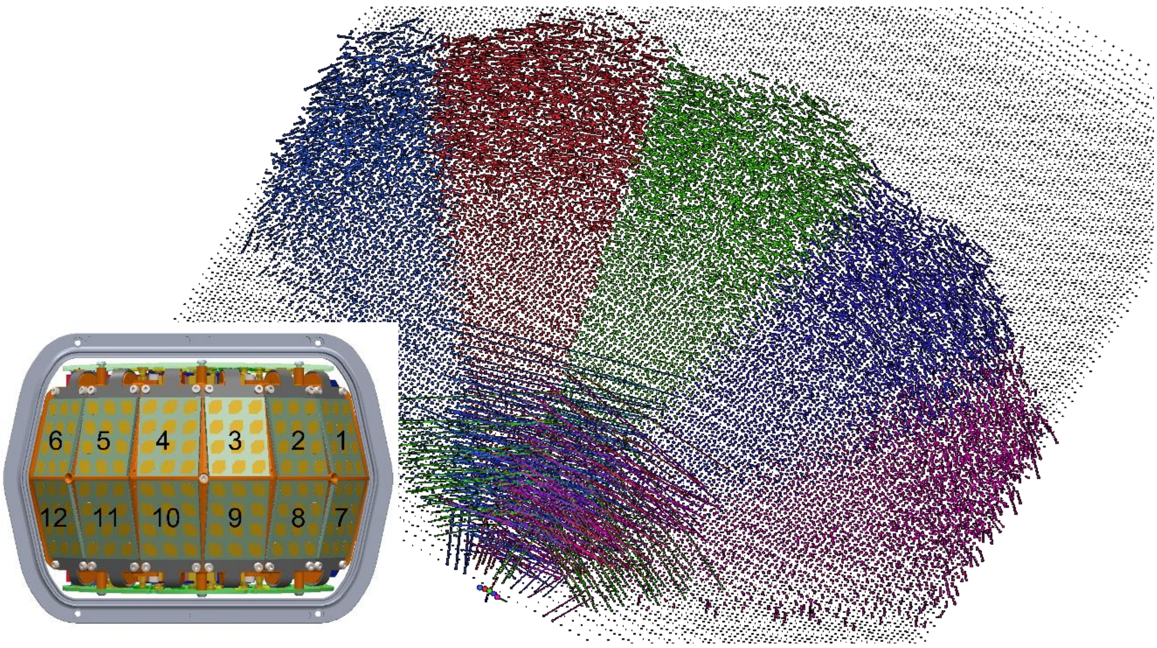

Figure 3 – Measurement space for DAA angular detection testing

Due to system sampling rates and communication bandwidths it is not possible to cover the entire search-volume instantaneously in range with a single chirp configuration profile[1], at the required resolution. It is therefore necessary to construct a set of profiles that when cycled through (at speed), can meet the full requirements of the *Radio Technical Commission for Aeronautics'* (RTCA's) DO-366B - *Minimum Operational Performance Standards* (MOPS) for Air-to-Air Radar for Traffic Surveillance.

For testing purposes, these profiles will be tested both individually and in combination, to better characterise the system. These profiles are designed to provide a minimum two-target resolution[2] of better than 12.5m (41ft) across a certain range-window but are also capable of detecting unique targets over a wider range.

It should be noted that the parameters in the tables below represent a very small subset of the available configuration settings. The full configuration set for each experiment is logged at runtime and will form part of the associated test report. More configurations may also be added over time to provide new capability.

3.2.1 Chirp Configuration Profile DAA-M1 (Very Short Range Interrupted FM)

Table 2 - Chirp Configuration Profile DAA-M1

Parameter	**Value**	**Units**	**Notes**
Setting			
Swept-frequency bandwidth	160	MHz	Full spectral bandwidth
Linear-chirp time	19.2	μs	Period of ADC sampling
# sub-chirps (on *and* off)	4	-	2 'cycles' of on/off PA switching
Sub-chirp time (on *or* off)	4.8	μs	PA active period per cycle
ADC sample-rate	40	MSPS	I/Q sample-rate @ 16-bit
# ADC samples	768	-	Samples over full sampling period
Full Tx/Rx Period	50	μs	Time between subsequent measurements.
Derived			
Minimum detection range	0	m	FMCW minimum range
Maximum detection range	720	m	Range-determined by chirp config
Minimum hi-res range	240	m	Range-determined by Rx-template overlap
Maximum hi-res range	720	m	Range-determined by Rx-template overlap

3.2.2 Chirp Configuration Profile DAA-M2 (Short Range Interrupted FM)

Table 3 - Chirp Configuration Profile DAA-M2

Parameter	**Value**	**Units**	**Notes**
Setting			
Swept-frequency bandwidth	48	MHz	Full spectral bandwidth
Linear-chirp time	19.2	μs	Period of ADC sampling
# sub-chirps (on *and* off)	2	-	1 'cycle' of on/off PA switching

[1] A 'chirp configuration profile' defines the specific swept-frequency slope, bandwidth, timing (including phase variation) and power for each radar transmit-receive measurement.
[2] Two-target resolution in this case is defined as the ability to discern to distinct peaks in the range-FFT, for two closely separated targets (in range) of equal received signal-strength.

Sub-chirp time (on *or* off)	9.6	µs	PA active period per cycle
ADC sample-rate	40	MSPS	I/Q sample-rate @ 16-bit
# ADC samples	768	-	Samples over full sampling period
Full Tx/Rx Period	50	µs	Time between subsequent measurements.
Derived			
Minimum detection range	0	m	FMCW minimum range
Maximum detection range	2400	m	Range-determined by chirp config
Minimum hi-res range	720	m	Range-determined by Rx-template overlap
Maximum hi-res range	2160	m	Range-determined by Rx-template overlap

3.2.3 Chirp Configuration Profile DAA-M3 (Medium Range Delayed FM)

Table 4 - Chirp Configuration Profile DAA-M3

Parameter	**Value**	**Units**	**Notes**
Setting			
Swept-frequency bandwidth	24	MHz	Full spectral bandwidth
Linear-chirp time	19.2	µs	Period of ADC sampling
# sub-chirps (on *and* off)	1	-	PA ‘on’ for Tx chirps only.
Sub-chirp time (on *or* off)	19.2	µs	PA active period per cycle
ADC sample-rate	40	MSPS	I/Q sample-rate @ 16-bit
# ADC samples	768	-	Samples over full sampling period
Full Tx/Rx Period	50	µs	Time between subsequent measurements.
Derived			
Minimum detection range	1200	m	Range-determined by Rx-template delay (24 µs)
Maximum detection range	6000	m	Range-determined by chirp config
Minimum hi-res range	2160	m	Range-determined by Rx-template overlap
Maximum hi-res range	5040	m	Range-determined by Rx-template overlap

3.2.4 Chirp Configuration Profile DAA-M4 (Long Range Delayed FM)

Table 5 - Chirp Configuration Profile DAA-M4

Parameter	**Value**	**Units**	**Notes**
Setting			
Swept-frequency bandwidth	24	MHz	Full spectral bandwidth
Linear-chirp time	19.2	µs	Period of ADC sampling
# sub-chirps (on *and* off)	1	-	2 ‘cycles’ of on/off PA switching
Sub-chirp time (on *or* off)	19.2	µs	PA active period per cycle
ADC sample-rate	40	MSPS	I/Q sample-rate @ 16-bit
# ADC samples	768	-	Samples over full sampling period
Full Tx/Rx Period	66	µs	Time between subsequent measurements.
Derived			
Minimum detection range	4080	m	Range-determined by Rx-template delay (24 µs)

Maximum detection range	8880	m	Range-determined by chirp config
Minimum hi-res range	5040	m	Range-determined by Rx-template overlap
Maximum hi-res range	7920	m	Range-determined by Rx-template overlap

These chirp profiles can then be combined to cover the entire region of interest for DAA, or, by varying the time between transmit and receive chirps, extended to operate at arbitrary range at a preconfigured resolution.

Note, the Radar Closest Performance Range (RCPR) defined in DO366-B for a class A2 radar, is 2200 feet (or 670m), whilst the maximum Radar Declaration Range (for a large intruder), again for a class A2 radar, is 4.59 Nautical Miles (or 8500m). The radar has therefore been designed to meet these performance criteria, which can be achieved using modes 2, 3 and 4, Mode 1 is mainly used for ground-based testing.

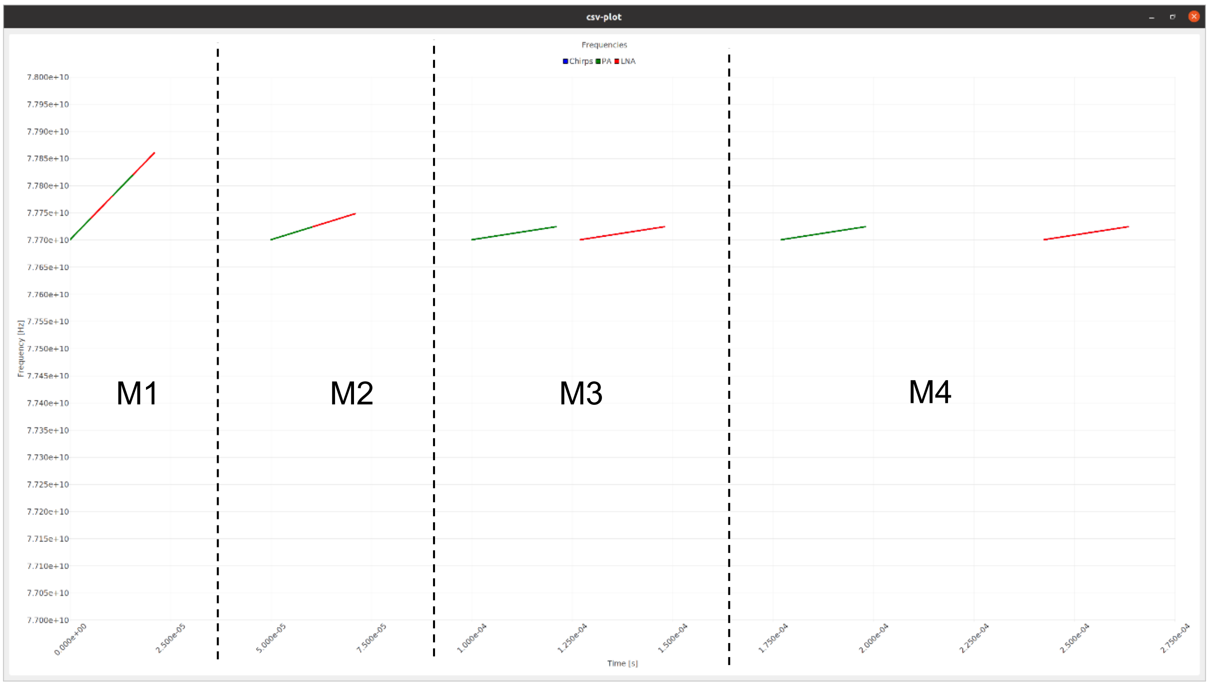

Figure 4 – M1 – M4 chirp profiles

3.3 Test hardware architecture

In addition to the SUT (two Boobook-DAA radar modules), a number of supporting elements are required as laid out in the architecture diagram of Figure 5. The Radar Payload itself is shown in Figure 6. This is built in a modular frame of extruded aluminium on a shock-mounted strut below the M1200 drone.

An NVIDIA Orin AGX module provides the processing system, to coordinate synchronised data-logging of the navigation system, gimbal-stabilised *electro-optic* (EO) and *infra-red* (IR) camera, laser-altimeter and two Boobook-DAA modules.

Each Jetson device (including Xavier NX modules on each radar module) has its own Solid State Drive (SSD) for storing sensor data at high-speed. It should be noted that when the radar is processing data at full speed (10 frames of full-volume data per second), it is not possible to log raw-data at that speed. There is therefore an engineering mode for capturing raw data separate to the online DAA mode will also be used.

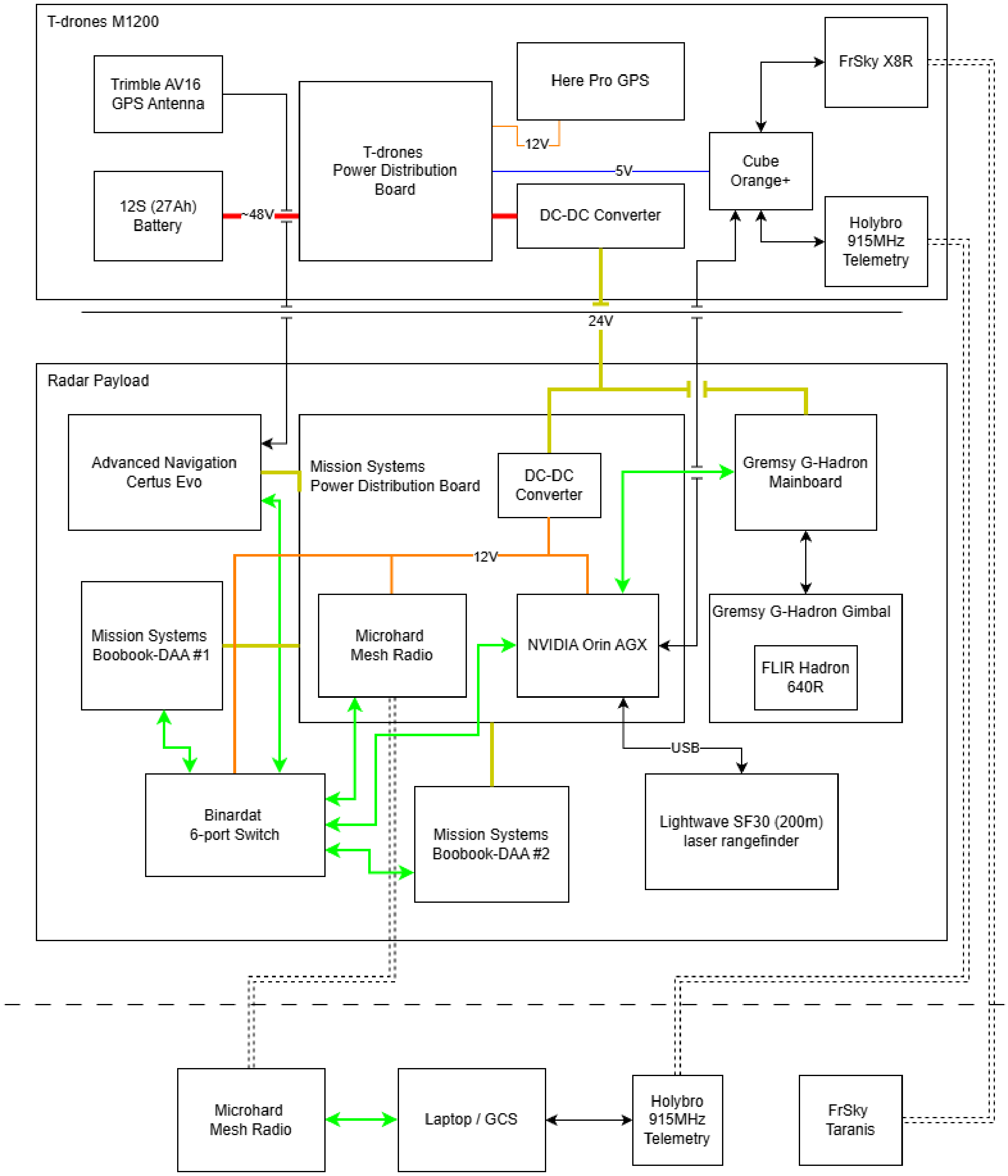

Figure 5 – Radar Payload Architecture

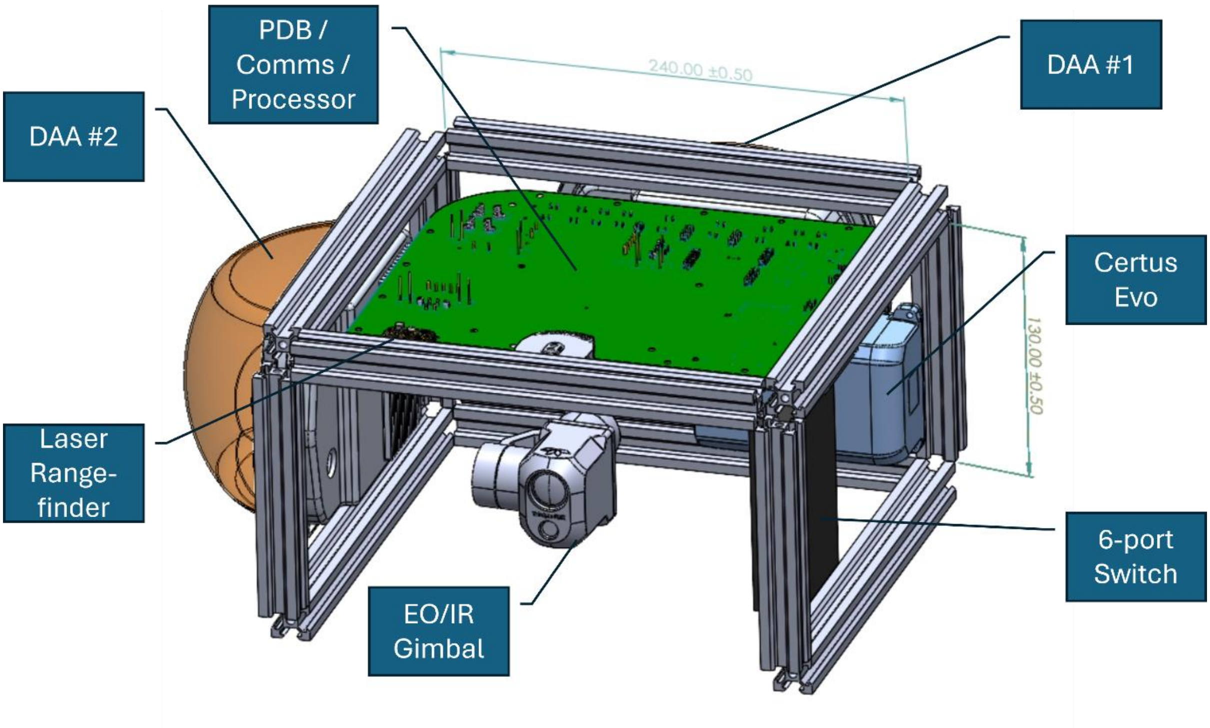

Figure 6 – Radar Payload Hardware Layout

3.3.1 The T-drones M1200 Host Platform

The platform to be used for the Boobook-DAA flight test campaign is a T-drones M1200 sub-25kg UAS, equipped with a Cube Orange+/Here-Pro flight system as seen in Figure 7. The radar payload, weighing ~3kg, will be mounted under the body of the aircraft.

Two DAA radar modules will be mounted at 90 degrees to one another (as they are nominally intended to be on an operational aircraft) in order to cover the full +/-110˚ field of regard. Note, this does mean that the 'forward direction' is actually at 45˚ with respect to the platform body, but as a quadcopter this is not particularly relevant. It also means that targets in that direction are actually the hardest to track in angle, requiring measurements from both radars to be compared in order to illicit the most accurate bearing information. In theory there should be sufficient azimuth coverage from each module to get two good measurements and compare, but this has yet to be tested in the real world.

Figure 7 – M1200 Experimental Test-Bed

3.3.2 M1200 Target Aircraft

The primary target aircraft will be a 2nd T-drones M1200 aircraft, which may be similarly equipped as the first with a radar payload (depending on component availability), or used purely as an airborne non-cooperative target.

Provisional trial dates of May 25th – June 5th have also been identified when multiple targets of opportunity will be in the vicinity of Cultana undertaking parallel trials for JDI.

3.4 Test Location

The test location will be within the Cultana Defence Practice Area in South Australia (exact location to be identified). JDI are awaiting feedback from DSO in relation to securing spectrum authorisation in this area, which will likely be tied to a specific geographic location.

For DAA test operations, the only pre-requisite is that both Host and Target platforms are able to maintain communication with the ground station and/or one another by mesh-radio and (under Mission Systems REoC) pilots operate within Visual Line Of Sight (VLOS) of their respective drone. While the maximum-range test metric requires a separation of up to 9km, it is likely (based on measurements made to date), that our communications system may encounter difficulties between 5 and 7km depending on the relative orientation of antennas.

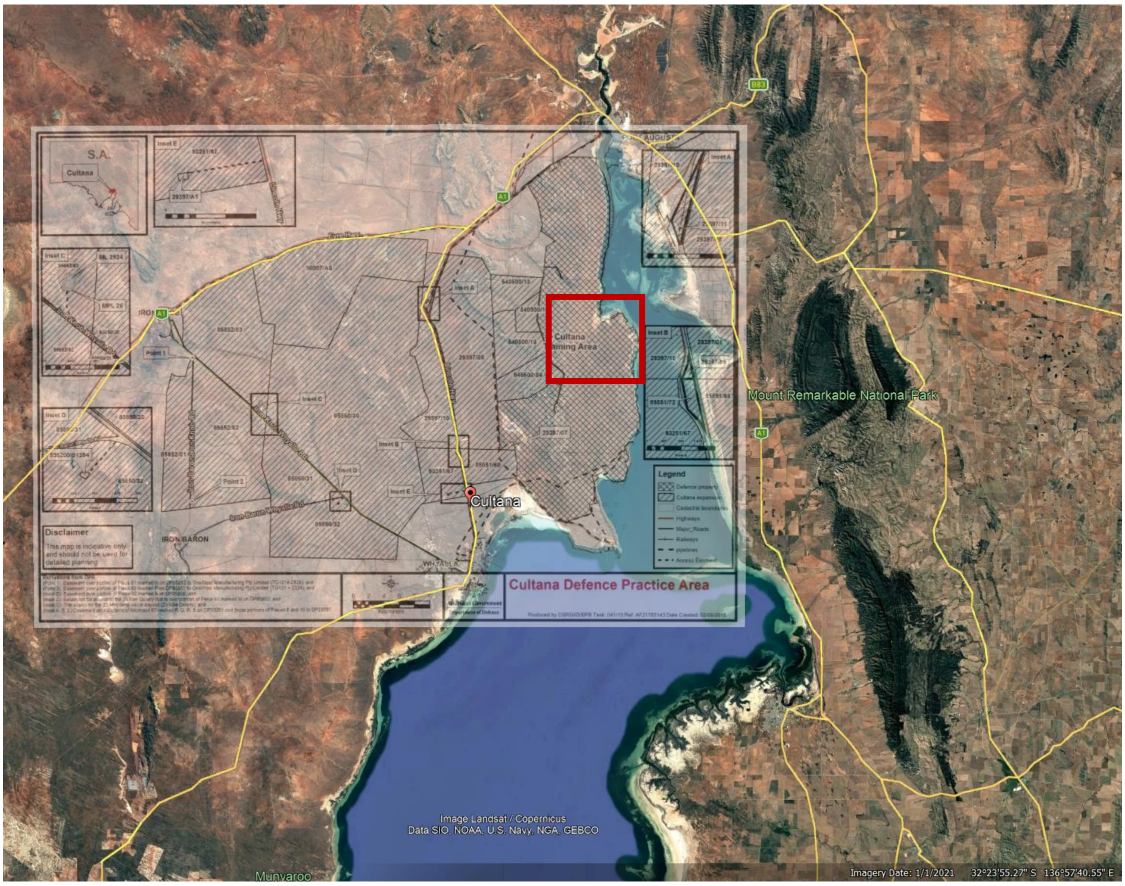

Figure 8 – Cultana Test Environment

Figure 9 – Prospective test geometry

Not having been to Cultana before, we do not have good familiarity with the area and where facilities might be located. We are therefore going off images from Google Earth, such as in Figure 8 and Figure 9, to provide representative test geometry. The key features required are:

1. Proximity to power to enable charging of flight batteries and ground control equipment. (Toilets, shade and water would also be useful.)
2. Proximity to a long straight road allowing the target platform and pilot to be easily repositioned at multiple ranges between 500m and 9km.
3. Absence of high terrain and other obstructions (e.g. power lines) between host and target platforms.

4 Test Procedures

The purpose of this section of the document is to describe the flight-test campaign in sufficient detail, to provide confidence that the envelope of the radar system's operational parameters can be sufficiently examined.

It is expected that 3-4 days of flight-testing will be required to complete the requisite experiments multiple times to characterise the performance of the radar.

During each experiment, limited telemetry will be available from each sensor to ensure the validity of the logging process. Following each experiment, it will therefore be necessary to directly access the radar payload via its external ethernet data-port to extract data and briefly verify the data collected.

Full analysis of this data will then be conducted after each day's trial activities have been completed.

4.1 Serial 0 - Ground DAA Tests

Operating in Cultana will be the first time this version of the radar will operate over the full spectrum in an outdoor environment (due to the spectral band it operates within proscribing outdoor use without a license). It is therefore the first opportunity to test long-range performance. Initial tests against airborne targets will therefore be made from the ground, with the radar payload detached from the test platform, and the target platform operated at close range (250-1000m).

Table 6 – Serial 0 – Ground Test DAA Experiments

Test ID	Short Title	Aim
0.1	Test comms and logging	Test that we can talk to (and log data from) everything inside the radar payload, and the target drone.
0.2	DAA-M1 profile acquisition	Very-short-range (250-900m) interrupted FM (full coverage) data collection. Target drone in the air at appropriate range and altitude.
0.3	DAA-M2 profile acquisition	Short-range (700-1300m) interrupted FM (full coverage) data collection. Target drone in the air at appropriate range and altitude.

During experiments 0.2 & 0.3, the flight profile of the target drone should be arranged such that at least two portions of flight provide the maximum doppler-velocity signature towards and away from the radar, whilst additional portions of the flight profile present a reduced doppler signature.

The exact flight-location of the target aircraft may vary depending on local conditions, noting that the aircraft must not directly over-fly personnel on the ground. The aircraft will also be flown under VLOS be one of Mission Systems' accredited remote pilots.

Pilot location and proximity to the aircraft will vary depending on operating altitude, based on a 1:1 rule, i.e. for a drone set to hover at 100m altitude, the pilot (and other personnel) will be positioned 100m away in ground-range.

4.2 Serial 1 - Aerial Tests (No Airborne Targets)

Following successful completion of Serial 0, Serial 1 will involve acquisition of data whilst the M1200 host platform is in the air. During this first phase of flight tests, no specific airborne targets will be operated, although it is possible that some targets of opportunity (including bats/birds) may present themselves. These experiments are primarily designed to measure the ground-clutter from the radar at various altitudes and configuration settings.

The following experiments described in Table 7 are expected to be conducted during a single flight, although repeat flights may be required to overcome any teething problems.

For experiments [1.2 – 1.6] a *raw-data* measurement repetition rate (MRR) of 500Hz has been chosen, as a compromise between data logging rate and sampling bandwidth, with ground-truth sensors being triggered once every 50 radar measurements, or 10FPS. Each experiment will record data for 3 minutes.

During each experiment the test-platform will move to a 3-D waypoint set in QGroundControl and hover there stationary for the duration of that test.

Table 7 – Serial 1 – Aerial Baseline Tests (No Airborne Targets)

Test ID	Short Title	Aim
1.1	Ground-truth sensor test (radar transmissions disabled)	Log navigation (including time-synchronisation), camera and lidar.
1.2	DAA-M1 chirp-profile acquisition	Very-short-range (250-900m) interrupted FM data collection. No target. 500Hz (min) MRR. 100m altitude. Rotating/yawing in hover at 3 RPM.
1.3	DAA-M2 Profile Acquisition	Short-range (720-2160m) interrupted FM data collection. No target. 500Hz (min) MRR. 200m altitude. Rotating/yawing in hover at 3 RPM.
1.4	DAA-M3 Profile Acquisition	Medium-range (2160-5040m) delayed FM data collection. No target. 500Hz (min) MRR. 300m altitude. Rotating/yawing in hover at 3 RPM.
1.5	DAA-M4 Profile Acquisition	Long-range (5040-7920m) delayed FM data collection. No target. 500Hz (min) MRR. 300m altitude. Rotating/yawing in hover at 3 RPM.
1.6	DAA-M1-4 Profile Acquisition	Multi-range (240m -7920m) data collection. No target. 500Hz (min) MRR. 300m altitude. Rotating/yawing in hover at 3 RPM.

4.3 Serial 2 - Aerial DAA Tests (With Slow-Moving Airborne Target)

This serial will require the coordinated flight of two M1200 drones, operated by their respective remote pilots. As before, flights will be conducted with a minimum pilot/drone separation based on the 1:1 rule.

Note, Mission Systems pilots are currently limited to VLOS operations (for sub-25kg UAS). These flights require drones to operate at up to 450m altitude, which will require a NOTAM to be in place, and ideally to be operated under BVLOS conditions. (We are actively looking at obtaining BVLOS training for one or more of our pilots, but a DRePL qualified defence pilot may be required.)

Table 8 – Serial 2 – Aerial DAA Tests (With Airborne Target)

Test ID	Short Title	Aim
2.01	DAA-M1-4 Profile – Target Location A (low alt)	As expt 1.6. Circling (5 m/s) target at 150m altitude, 50m radius, 500m +/-50m ground-range.
2.02	DAA-M1-4 Profile – Target Location B (low alt)	As expt 1.6. Circling (5 m/s) target at 150m altitude, 50m radius, 1000m +/-50m ground-range.
2.03	DAA-M1-4 Profile – Target Location C (low alt)	As expt 1.6. Circling (5 m/s) target at 150m altitude, 50m radius, 2000m +/-50m ground-range.
2.04	DAA-M1-4 Profile – Target Location D (low alt)	As expt 1.6. Circling (5 m/s) target at 150m altitude, 50m radius, 4000m +/-50m ground-range.
2.05	DAA-M1-4 Profile – Target Location E (low alt)	As expt 1.6. Circling (5 m/s) target at 150m altitude, 50m radius, 8000m +/-50m ground-range.
2.06	DAA-M1-4 Profile – Target Location A (same alt)	As expt 1.6. Circling (5 m/s) target at 300m altitude, 50m radius, 500m +/-50m ground-range.
2.07	DAA-M1-4 Profile – Target Location B (same alt)	As expt 1.6. Circling (5 m/s) target at 300m altitude, 50m radius, 1000m +/-50m ground-range.
2.08	DAA-M1-4 Profile – Target Location C (same alt)	As expt 1.6. Circling (5 m/s) target at 300m altitude, 50m radius, 2000m +/-50m ground-range.
2.09	DAA-M1-4 Profile – Target Location D (same alt)	As expt 1.6. Circling (5 m/s) target at 300m altitude, 50m radius, 4000m +/-50m ground-range.
2.10	DAA-M1-4 Profile – Target Location E (same alt)	As expt 1.6. Circling target at (5 m/s) 300m altitude, 50m radius, 8000m +/-50m ground-range.
2.11	DAA-M1-4 Profile – Target Location A (high alt)	As expt 1.6. Circling (5 m/s) target at 450m altitude, 50m radius, 500m +/-50m ground-range.
2.12	DAA-M1-4 Profile – Target Location B (high alt)	As expt 1.6. Circling (5 m/s) target at 450m altitude, 50m radius, 1000m +/-50m ground-range.
2.13	DAA-M1-4 Profile – Target Location C (high alt)	As expt 1.6. Circling (5 m/s) target at 450m altitude, 50m radius, 2000m +/-50m ground-range.
2.14	DAA-M1-4 Profile – Target Location D (high alt)	As expt 1.6. Circling (5 m/s) target at 450m altitude, 50m radius, 4000m +/-50m ground-range.
2.15	DAA-M1-4 Profile – Target Location E (high alt)	As expt 1.6. Circling (5 m/s) target at 450m altitude, 50m radius, 8000m +/-50m ground-range.

If during these experiments it is determined that performance is worse (or better) than expected, target locations may be adjusted as required to identify the true performance limits of the radar system.

4.4 Serial 3 - Aerial DAA Tests (Multiple Airborne Targets of Opportunity)

JDI have indicated that during the period of May 25th – June 5th 2026 there may be multiple trials underway within the Cultana Training Area that may allow for measurements against airborne targets of opportunity. Whilst insufficient information is currently available to make a detailed experimental plan, there are certain statements that can be made as to what this might entail:

1. These operations must be coordinated and deconflicted to ensure that there is no possibility of air-to-air incidents, particularly in proximity to personnel on the ground. A no-fly-zone of 500m from the stationary hover position of the DAA test platform (at a nominal 300m altitude) would likely be sufficient.
2. The flight-trajectory of such targets must be known, and recorded, so that true targets can be positively identified from false ones. Ideally, this means obtaining timestamped ground-truth tracks from the targets in question (after flight).
3. The location and timing of such flights should be known in advance (where to within a few 100m, and when within a few minutes), with sufficient notice to coordinate and implement a DAA test-platform flight-plan.

5 Safety Considerations

This section summarises the risks associated with the trial as specified above.

5.1 Aircraft Operations

The most important considerations during the trial must be to personnel safety, both for those read into the trial and for other air users.

In general, Mission Systems' *standard operating procedures* (SOPs) for conduction flight operations will be adhered to at all times.

5.1.1 Radar Power Level

Mission Systems will undertake a risk assessment for the radar modules installed on the radar-payload following the *Australian Radiation Protection and Nuclear Safety Agency* (ARPANSA) standard for limiting exposure to radio-frequency fields (between 100kHz and 300GHz).

As a preventative measure, a software-based safety cut-out based on measured altitude from the INS will be used to limit activation of the radar modules under all circumstances except for Serial 0 (Ground DAA tests).

5.1.2 Batteries

The radar-payload is a self-contained unit requiring only 24V input. This is provided from the M1200 12S Lithium-Ion batteries and a suitably rated DC-DC converter, or from a benchtop power supply as required.

The Mission Systems' SOP for Charging, Storing, and Handling Lithium (Li) Batteries should be followed when handling these batteries. The minimum precaution is that battery safe enclosures should be used on the ground, and F-500 fire extinguishers and fire blankets should be available at both Host and Target platform take-off/landing sites.

5.2 Drone Operations

The M1200 is included in Mission Systems RPAS Operations Register, which is a live document stored on google drive.

1. As a sub-25kg RPAS, the M1200 carries a lot of kinetic energy and must therefore be treated as posing a significant threat to personnel and the environment in its own right. Careful adherence to its operations manual should therefore be followed at all times.
2. For the purposes of providing good ground-truth, both the host and target M1200 should be flown on a preset (automated) flight path. However, the RPAS pilots must always have the ability to manually take control of the aircraft, particularly to ensure deconfliction with other aircraft in the environment.

6 Verification Test Procedure for Post-Flight Analysis

As per Table 1, there are a number of TPMs that can be verified by direct measurement, either using a Spectrum Analyser, scales or tape-measure. Others require significantly more effort to ensure that the requirements derived from DO-366B can be met.

We have taken the view that there are a core subset of the DO-366 requirements (and hence those listed in the System Specification document) which would indicate whether the radar is fundamentally on the right track with regards to product development. If these can be met, everything else is a matter of tinkering with software around the edges. It comes down to the coverage, accuracy and latency of tracked targets in range and angle.

Targets will be considered successfully 'tracked-online' if the output of the detection/tracking filter produces a consistent target track over consecutive track reports (at a rate of 1Hz).

Table 9 – Tracker accuracy requirements (from DO-365B)

Parameter	Mean	Standard Deviation	95% Containment	RMSE
Range Error (feet (ft))	50	70	±166	86
Range Rate Error (feet per second) (fps) Class A	8	10	±24.5	12.8
Azimuth Angle Error (Degrees) Class A	0.5	1	±2.2	1.1
Elevation Angle Error (Degrees) Class A	0.5	1	±2.2	1.1

Whilst working up to full online-tracking (serials 0 and 1), raw data will be acquired at a lower repetition rate, and this will be used to tune tracking filters in range, bearing and doppler (range-rate). Analysis of this raw data may also be used to validate certain parameters.

In general, these measurements will also require that appropriate time-stamped ground-truth state information (position and speed) is obtained from the target aircraft to ensure that radar measurements are accurate.

6.1 TPM-11.2.15D - Minimum range (single module) < 500m

Experiments 2.01, 2.06 and 2.11 are designed to track targets at 500m +/- 50m range at relative altitudes of -150, 0 and +150m respectively.

6.2 TPM-11.2.3A - Maximum range (single module)

The remaining experiments in Serial 2 will be used to determine the effective maximum range of the radar, out to a target range of 8km.

6.3 TPM-12.5.1A / TPM-11.2.16D - DAA target tracking error in Azimuth / DAA-mode azimuth field of regard (two modules)

As the host platform will be performing continual rotations whilst target-tracking measurements are made, the full azimuth coverage of both radar modules will be profiled, including the cross-over as a target passes from one module to the other. This profile will also allow the monopulse angular error to be measured (after appropriate calibration) over each pair of sub-arrays.

6.4 TPM-11.2.3A / TPM-11.2.17D - DAA target tracking error in Elevation / DAA-mode elevation field of regard (two modules)

At the minimum target range of 450m, this corresponds to minimum and maximum elevation angles of +/-18˚. As for azimuth, as the target moves in range, the angular error may be calculated (after calibration). It is worth noting that that the relative change in elevation during any single experiment will be quite small, e.g. the difference between a target at 450 and 550m range (and 150m altitude offset) is only 3˚.

6.5 TPM-11.2.18D - DAA-mode target tracking error in range

As for previous tests, the output of the tracker must be compared to ground-truth target track to determine the mean and standard deviation in range error for a particular target. This may also vary depending on radar mode (M1 - M4) hence should be calculated for each target location (A-E).

6.6 TPM-11.2.19D - DAA-mode target tracking error in range-rate

Calculation of ground-truth range-rate is difficult for an arbitrary target, but for a circling target should be obvious as a sinusoidal output from the doppler tracker. Again, it should be noted that the maximum relative-velocity of the M1200 target aircraft is very low (particularly as the host platform will itself be hovering). There may therefore be very little doppler to detect as it will only reach maxima (5 m/s relative velocity) at two points on the orbit. For this particular test it may be that other airborne targets of opportunity (Serial 3) are more relevant for producing sufficient Doppler to provide useful data.